Further Integration Techniques

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Integration by substitution**

Evaluate $\int 2x(x^2+1)^4\,dx$ using the substitution $u = x^2 + 1$.

$du = 2x\,dx$. So $\int (x^2+1)^4 \cdot 2x\,dx = \int u^4\,du = u^5/5 + C = (x^2+1)^5/5 + C$.

2 **[Foundational]** • **Target: Integration by parts**

Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = (x-1)e^x + C$.

3 **[Foundational]** • **Target: Repeated integration by parts**

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

First application: $u = x^2$, $dv = e^x dx$: $\int x^2e^x\,dx = x^2e^x - \int 2xe^x\,dx$. From the previous problem, $\int xe^x\,dx = (x-1)e^x + C$, so $\int 2xe^x\,dx = 2(x-1)e^x$. Result: $x^2e^x - 2(x-1)e^x + C = (x^2 - 2x + 2)e^x + C$.

4 **[Foundational]** • **Target: Integration using trigonometric identities**

Evaluate $\int \cos^2 x\,dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.

$\int (1 + \cos 2x)/2\,dx = 1/2(x + \sin 2x/2) + C = x/2 + \sin 2x/4 + C$.

5 **[Foundational]** • **Target: Integration using partial fractions**

Evaluate $\int 1/(x^2-1)\,dx$ by first expressing the integrand in partial fractions.

$1/(x^2-1) = 1/((x-1)(x+1)) = A/(x-1) + B/(x+1)$. Solving: $A = 1/2$, $B = -1/2$.

$\int (1/2/(x-1) - 1/2/(x+1))dx = 1/2\ln|x-1| - 1/2\ln|x+1| + C = 1/2\ln|(x-1)/(x+1)| + C$.

6 **[Foundational]** • **Target: Integration involving inverse trigonometric results**

Evaluate $\int 1/(1+x^2)\,dx$, and use it to evaluate $\int_0^1 1/(1+x^2)\,dx$.

$\int 1/(1+x^2)\,dx = \arctan x + C$. So $\int_0^1 1/(1+x^2)\,dx = \arctan(1) - \arctan(0) = \pi/4 - 0 = \pi/4$.

7 **[Foundational]** • **Target: Integration by substitution with trigonometric functions**

Evaluate $\int \sin^3 x\,dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.

Rewrite: $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x\,dx$.

$\int \sin x(1 - \cos^2 x)\,dx = -\int (1 - u^2)\,du = -u + u^3/3 + C = -\cos x + \cos^3 x/3 + C$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Integration using a Weierstrass-style substitution**

Evaluate $\int 1/(5+4\cos x)\,dx$ using the substitution $t=\tan(x/2)$, given that this substitution transforms $\cos x=(1-t^2)/(1+t^2)$ and $dx=2/(1+t^2)dt$, leading to the simplified integrand $2/(9+t^2)$.

Substituting: $\int 1/(5+4(1-t^2)/(1+t^2))\cdot 2/(1+t^2)\,dt=\int 2/(5(1+t^2)+4(1-t^2))\,dt=\int 2/(9+t^2)\,dt$. Using the standard result $\int 1/(a^2+t^2)\,dt=(1/a)\arctan(t/a)+C$ with $a=3$: $2\cdot(1/3)\arctan(t/3)+C=(2/3)\arctan((1/3)\tan(x/2))+C$.

9 **[Intermediate]** • **Target: Integration by parts with a trigonometric-exponential product**

Evaluate $\int e^x\cos x\,dx$, applying integration by parts twice and solving the resulting equation for the original integral.

Let $I=\int e^x\cos x\,dx$. First application: $u=\cos x$, $dv=e^x dx$, so $du=-\sin x\,dx$, $v=e^x$: $I=e^x\cos x+\int e^x\sin x\,dx$. Second application on $\int e^x\sin x\,dx$: $u=\sin x$, $dv=e^x dx$: $\int e^x\sin x\,dx=e^x\sin x-\int e^x\cos x\,dx=e^x\sin x-I$. Substituting back: $I=e^x\cos x+e^x\sin x-I$, so $2I=e^x(\cos x+\sin x)$, giving $I=e^x(\cos x+\sin x)/2+C$.

10 **[Intermediate]** • **Target: Finding the area between two curves using integration**

Find the area of the region enclosed between $y=\sin x$ and $y=\cos x$ for $\pi/4\le x\le 5\pi/4$ (the interval on which $\sin x\ge\cos x$).

Area $=\int_{\pi/4}^{5\pi/4}(\sin x-\cos x)\,dx=[-\cos x-\sin x]_{\pi/4}^{5\pi/4}$. At $x=5\pi/4$: $-\cos 5\pi/4-\sin 5\pi/4=\sqrt{2}/2+\sqrt{2}/2=\sqrt{2}$. At $x=\pi/4$: $-\cos\pi/4-\sin\pi/4=-\sqrt{2}/2-\sqrt{2}/2=-\sqrt{2}$. Area $=\sqrt{2}-(-\sqrt{2})=2\sqrt{2}$.

11 **[Intermediate]** • **Target: Finding the area between two curves using integration**

A particle's velocity is given by $v(t)=t\cos t$ metres per second. Find the particle's displacement over the interval $0\le t\le\pi$, using integration by parts.

Displacement $=\int_0^{\pi} t\cos t\,dt$. Using integration by parts with $u=t$, $dv=\cos t\,dt$, so $du=dt$, $v=\sin t$: $\int t\cos t\,dt=t\sin t-\int\sin t\,dt=t\sin t+\cos t+C$. Evaluating from 0 to π: $[\pi\sin\pi+\cos\pi]-[0\sin 0+\cos 0]=[0-1]-[0+1]=-1-1=-2$. Displacement $=-2$ metres.

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12 **[Intermediate]** • **Target: Finding the area between two curves using integration**

This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int x/(x^2+4)\,dx$. (b) Use integration by parts to find $\int \arctan x\,dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1+x^2)dx = \arctan x + C$, evaluate $\int_0^1 \arctan x\,dx$, giving an exact answer in terms of π and $\ln$.

(a) $u = x^2+4$, $du = 2x\,dx$: $\int x/(x^2+4)\,dx = 1/2\int 1/u\,du = 1/2\ln|x^2+4| + C$. (b) Let $u = \arctan x$, $dv = dx$, so $du = (1/(1+x^2))\,dx$, $v = x$: $\int \arctan x\,dx = x\arctan x - \int x/(1+x^2)\,dx = x\arctan x - 1/2\ln(1+x^2) + C$ (using the result from part (a), with 4 replaced by 1). (c) Evaluating from 0 to 1:

$[1\cdot\arctan 1 - 1/2\ln 2] - [0 - 1/2\ln 1] = \pi/4 - 1/2\ln 2$.

13 **[Intermediate]** • **Target: Reduction formulas**

Given the reduction formula $I_n = \int x^n e^x\,dx = x^n e^x - nI_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x\,dx$.

$I_1 = x^1 e^x - 1\cdot I_0 = xe^x - e^x$ (ignoring the constant until the final step).

$I_2 = x^2e^x - 2I_1 = x^2e^x - 2(xe^x - e^x) = x^2e^x - 2xe^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$, matching the result found earlier in this worksheet by direct repeated integration by parts.

14 **[Intermediate]** • **Target: Volume of revolution using integration**

The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis. Find the volume of the resulting solid, using $V = \pi\int_0^4 y^2\,dx$.

Since $y = \sqrt{x}$, $y^2 = x$. $V = \pi\int_0^4 x\,dx = \pi[x^2/2]_0^4 = \pi(8-0) = 8\pi$ cubic units.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Integration using the sine double-angle identity** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x-3}{x^2-2x-3}\,dx$

Factor the denominator: $x^2 - 2x - 3 = (x-3)(x+1)$. Set up partial fractions:

$\frac{5x-3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1} \Rightarrow 5x - 3 = A(x+1) + B(x-3)$ Using Heaviside cover-up: Set $x = 3$:

$5(3) - 3 = A(3+1) \Rightarrow 12 = 4A \Rightarrow A = 3$. Set $x = -1$: $5(-1) - 3 = B(-1-3) \Rightarrow -8 = -4B \Rightarrow B = 2$.

Integrate term-by-term.

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16 **[Challenging] • Target: Volume of revolution using integration • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Decompose the rational expression $R(x) = \frac{7x-1}{x^2-x-6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Set up the decomposition: $\frac{7x-1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x-1}{x+2}\Big|_{x=3} = \frac{7(3)-1}{3+2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x-1}{x-3}\Big|_{x=-2} = \frac{7(-2)-1}{-2-3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x-3} + \frac{3}{x+2}$.

17 **[Challenging] • Target: Substitution with a shifting definite integral bound • Cross-Lesson: 4.02 The Poisson Distribution**

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Find points of intersection: $4x - x^2 = x \Rightarrow 3x - x^2 = 0 \Rightarrow x(3-x) = 0$ The curves intersect at $x = 0$ and $x = 3$. Between 0 and 3, the curve $y = 4x - x^2$ lies above the line $y = x$. The enclosed area is: $A = \int_0^3 [(4x - x^2) - x]\,dx = \int_0^3 (3x - x^2)\,dx$ Integrate: $A = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \left(\frac{3(9)}{2} - \frac{27}{3}\right) - 0 = \frac{27}{2} - 9 = \frac{27-18}{2} = \frac{9}{2} = 4.5$

18 **[Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations**

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

Substitute $u = \ln x$. The derivative is $du = \frac{1}{x}\,dx$. The integral simplifies directly to: $\int \cos(u)\,du = \sin(u) + C = \sin(\ln x) + C$

19 **[Challenging] • Target: Integration by substitution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging] • Target: Integration by parts • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Choosing an appropriate integration technique

Which integration technique is most appropriate for evaluating $\int x^3 \sin x \, dx$?

a. Integration by parts, applied three times (reducing the power of x each time) ✓

b. Substitution $u = \sin x$

c. Partial fractions

d. The double-angle identity for $\sin$

Since the integrand is a product of a polynomial and a trigonometric function, integration by parts is appropriate — and because the polynomial factor is x^3 (not x^1), the process must be repeated three times, each time reducing the power of x by one, until the polynomial factor disappears.

22 [Foundational] • Target: Integration by parts

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?

a. 17 ✓

b. 7

c. 2.4

d. -7

The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\,dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\,dx$, is 12 square units. This implies that this region contributes positively to the net signe.

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23 [Foundational] • Target: Repeated integration by parts

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

24 [Foundational] • Target: Integration using trigonometric identities

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_{a}^{b}(5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. **8/3** ✓

d. 16/3

The region R is bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. We are revolving this region about the vertical line $x = 5$. We will use the method of cylindrical shells. The radius of a cylindrical shell at a given x is the distance from the axis of revolution $x = 5$ to x, which is $5 - x$. The height of the shell is the difference in the y-values of the bounding curves, which is $\sqrt{x} - 0 = \sqrt{x}$. The thickness of the shell is dx. The volume of a single cylindrical shell is.

25 [Foundational] • Target: Integration using partial fractions

The integral $\int_{0}^{1} e^{-x^2}dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$ ✓

c. $1 + 1/3 \approx 1.333$

d. 0

$\int_{0}^{1} e^{-x^2}dx \approx 1 - 1/3 + \cdots \approx 0.667.$

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26 [Foundational] • Target: Evaluating a definite integral

Integration by parts chooses u such that:

a. u is simple to integrate

b. **du/dx is simpler than u ✓**

c. dv is the whole integrand

d. u contains e^x always

Choose u so that du/dx simplifies the remaining integral (LIATE rule).

27 [Foundational] • Target: Integration involving inverse trigonometric results

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:

a. $f > g$ and $f < g$ alternates

b. $f > 0$ and $g < 0$

c. $f(x) \geq g(x)$ **on** $[a, b]$ ✓

d. always

Applying Applications of Integration principles, Area when $f \geq g$. Therefore, the verified correct option is $f(x) \geq g(x)$ on $[a, b]$.

28 [Foundational] • Target: Integration by substitution with trigonometric functions

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$ ✓

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

On $[0, \pi/2]$, $I_n = \frac{n-1}{n} I_{n-2}$ with $I_0 = \pi/2$ and $I_1 = 1$. Applying it repeatedly gives $\frac{5\pi}{32}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integration by parts with a logarithmic integrand

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$ ✓

d. $\pi(\sqrt{2} - 1)$

Applying Applications of Integration principles, Washer method. Therefore, the verified correct option is $\pi(2 - \pi/2)$.

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30 [Intermediate] • Target: Integration using a Weierstrass-style substitution

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π ✓

d. $\sqrt{\pi}$

$V = 2\pi\int_0^{\sqrt{\pi/2}} x\cos(x^2)dx = \pi[\sin(x^2)]_0^{\sqrt{\pi/2}} = \pi\sin(\pi/2) = \pi.$

31 [Intermediate] • Target: Integration by parts with a trigonometric-exponential product

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x\ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

32 [Intermediate] • Target: Finding the area between two curves using integration

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$ ✓

d. $\frac{7}{10}$

$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}.$

33 [Intermediate] • Target: Reduction formulas

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$ ✓

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

$S = 2\pi\int_{-R}^{R}\sqrt{R^2 - x^2}\sqrt{1 + x^2/(R^2 - x^2)}\,dx = 4\pi R^2.$

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34 [Intermediate] • Target: Integration by substitution with a linear inner function

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$ ✓

c. $\frac{4}{5}$

d. $\frac{9}{25}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$.

35 [Intermediate] • Target: Integration using the sine double-angle identity

Partial fractions: $\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$. Find A and B:

a. $A = 2, B = 1$ ✓

b. $A = 1, B = 2$

c. $A = 2, B = 3$

d. $A = 3, B = 2$

$2x + 3 = A(x+1) + B$; at $x = -1$: $B = 1$; coeffs of x: $A = 2$.

36 [Intermediate] • Target: Volume of revolution using integration

Volume integral: using spherical coordinates, $dV =$

a. $dr\,d\theta\,d\phi$

b. $r^2\sin\phi\,dr\,d\theta\,d\phi$ ✓

c. $r\sin\theta\,dr\,d\theta\,d\phi$

d. $r^2\,dr\,d\theta\,d\phi$

Spherical: $dV = r^2\sin\phi\,dr\,d\theta\,d\phi$.

37 [Intermediate] • Target: Substitution with a shifting definite integral bound

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

a. trig of t

b. **rational functions of t** ✓

c. exponentials of t

d. polynomials

Weierstrass: rationalises all trig expressions.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations

$\int \frac{1}{x(x-1)}\,dx$ by partial fractions:

a. $\ln|x| - \ln|x-1| + C$

b. $\ln|x(x-1)| + C$

c. $\frac{1}{x-1/2} + C$

d. $\ln|x-1| - \ln|x| + C$ ✓

$\frac{1}{x(x-1)} = \frac{-1}{x} + \frac{1}{x-1}$.

39 [Challenging] • Target: Integration by substitution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$.

Example for $f = e^x$:

a. gives nothing useful

b. yields: $\ln b - \ln a$

c. gives $e^b - e^a$

d. **yields:** $b\ln b - a\ln a - (b-a)$ ✓

Inverse function integration applied to $\ln$.

40 [Challenging] • Target: Integration by parts • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

$\int \frac{1}{x^3 - x}\,dx$: partial fractions decomposition is:

a. $\frac{A}{x^2+1} + \frac{B}{x-1}$

b. $\frac{A}{x^3} + \frac{B}{x}$

c. $\frac{Ax+B}{x^2} + \frac{C}{x-1}$

d. $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$ ✓

$x^3 - x = x(x-1)(x+1)$: three linear factors.

41 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 6.01 Matrices (AI HL)

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

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42 [Challenging] • Target: Integration using trigonometric identities • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$ ✓

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

$A = \frac{1}{2}\int (r_2^2 - r_1^2)d\theta$.

43 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Tabular integration (repeated parts): $\int x^3 e^x\,dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3 ✓

x^3 differentiates to 0 in 3 steps.

44 [Challenging] • Target: Evaluating a definite integral • Cross-Lesson: 1.01 Proof by Mathematical Induction

Partial fractions for $\frac{1}{(x+1)^2}$ uses:

a. **repeated linear factor** ✓

b. linear factors

c. irreducible quadratic

d. none

$\frac{A}{x+1} + \frac{B}{(x+1)^2}$.

45 [Challenging] • Target: Integration involving inverse trigonometric results • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

$\int \sin^4 x\,dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$ ✓

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

Double application of half-angle.

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